SPARELINE

DATANOVA 2026 / DN-OR-01

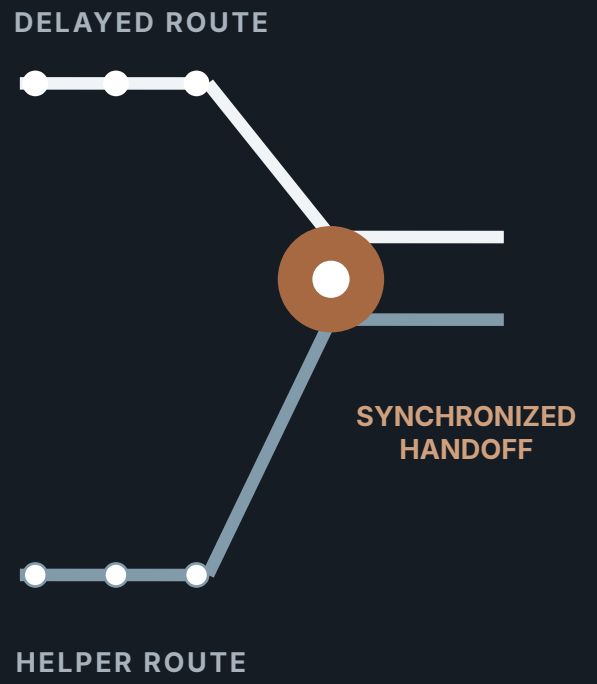
When a delivery route falls behind, know exactly what to move.

A dispatch decision tool for late-route recovery.

Timothy Popoola · Anu Kolapo

WORKING PRODUCT • SPARELINE-STEEL.VERCEL.APP

A rescue must work for both routes.



The hard decision starts after the plan fails.

One van is running late. Another driver could help, but the wrong rescue spreads the delay to a second route.

01

A route slips

Unexpected delay changes the remaining arrival times.

02

Someone offers to help

A second van has its own parcels, limits and commitments.

03

The dispatcher decides

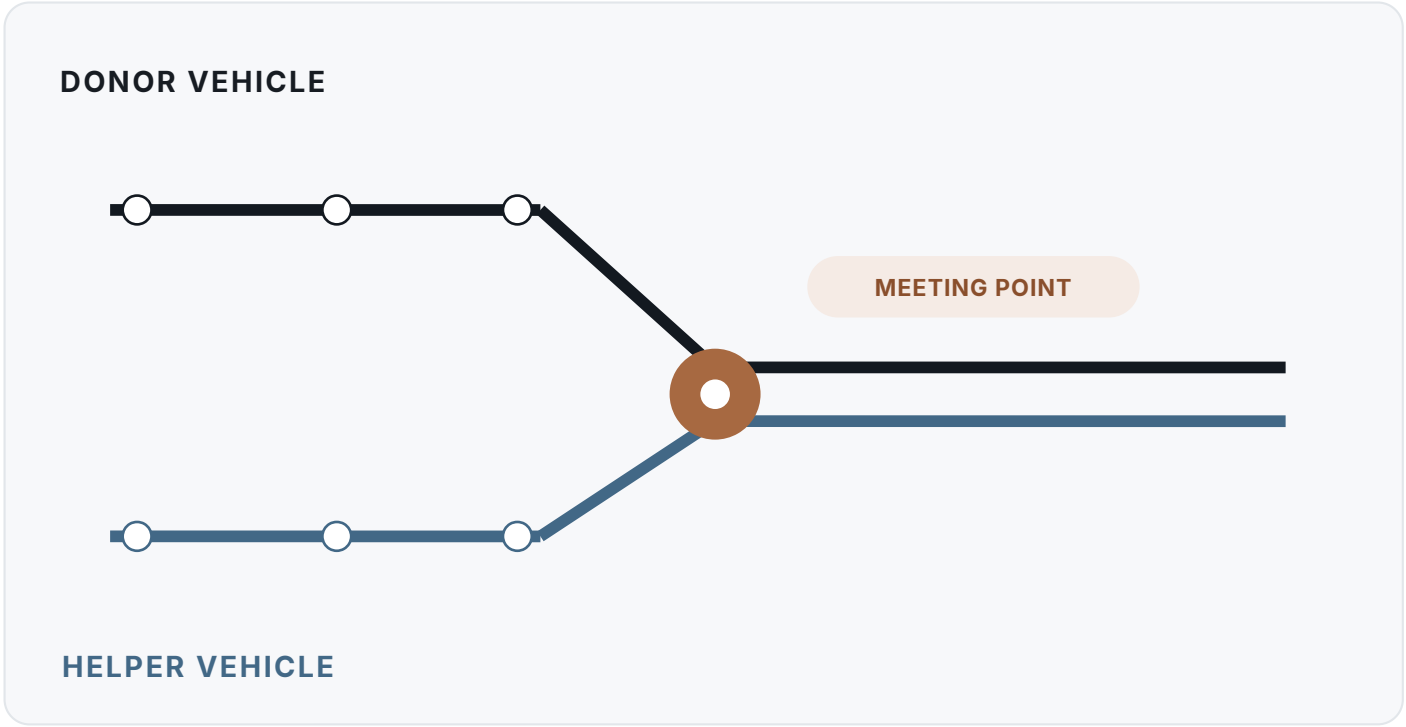
Which packages move, where do drivers meet, and who becomes late?

The question: **Can a rescue save the late route without breaking the helping route?**

Problem anchor: public accounts from delivery drivers sent on rescues, and delivery-operations research. Not customer interviews.

Rescue is a two-route decision.

Moving a parcel only helps if two vehicles can meet and still finish feasible schedules.



01 Capacity

Can the helper actually carry the transferred load?

02 Timing

Can both vans arrive at the same place?

03 Windows

Do customers still get parcels inside their delivery window?

04 Side effects

Does the helper now miss its own deliveries?

Diagram is illustrative. In the product, meeting time, handoff duration and both revised routes are calculated by the solver.

One rescue recommendation. All consequences visible.

Spareline turns a route disruption into an actionable, checkable handoff plan.

INPUT

01

Identify risk

Replay each van to now, apply the delay, and find the stops that will miss a window or the end of the shift.

OPTIMIZE

02

Evaluate transfers

Try each helper van, meeting point and group of stops. Check van space and both new routes.

OUTPUT

03

Issue instructions

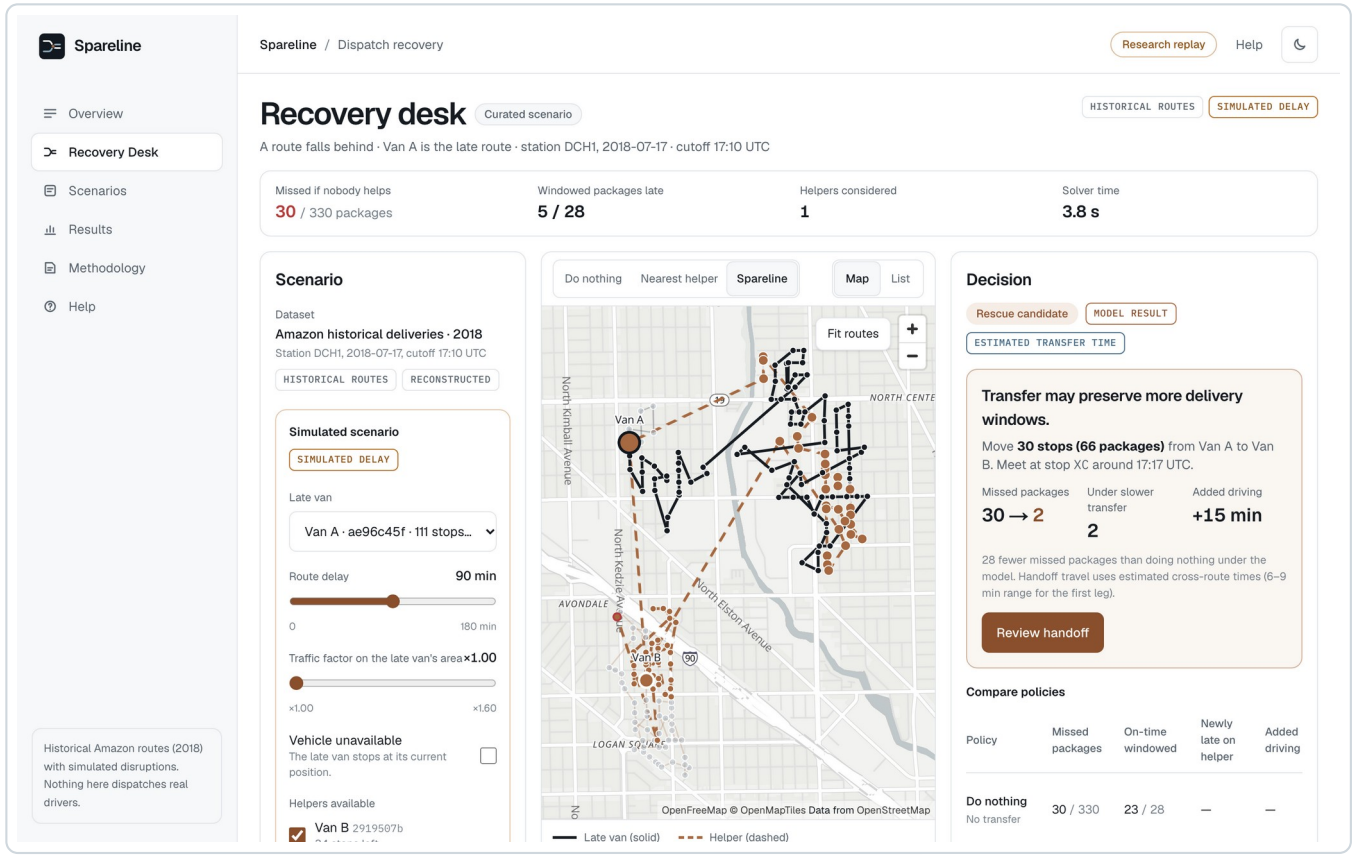
Compare with doing nothing and with the nearest van, then export both drivers' new stop lists.

It can also return "no safe rescue" or "low confidence": it withheld a transfer in 280 of 354 at-risk evaluation scenarios.

Designed for the dispatcher’s next action.

An operations desk, not a chatbot. The live optimization result drives every status and instruction.

REAL CAPTURE • spareline-steel.vercel.app • SCENARIO A



01 Map and decision from one run

Donor solid, helper dashed, meeting point and moved stops marked; stop table as fallback.

02 Three policies side by side

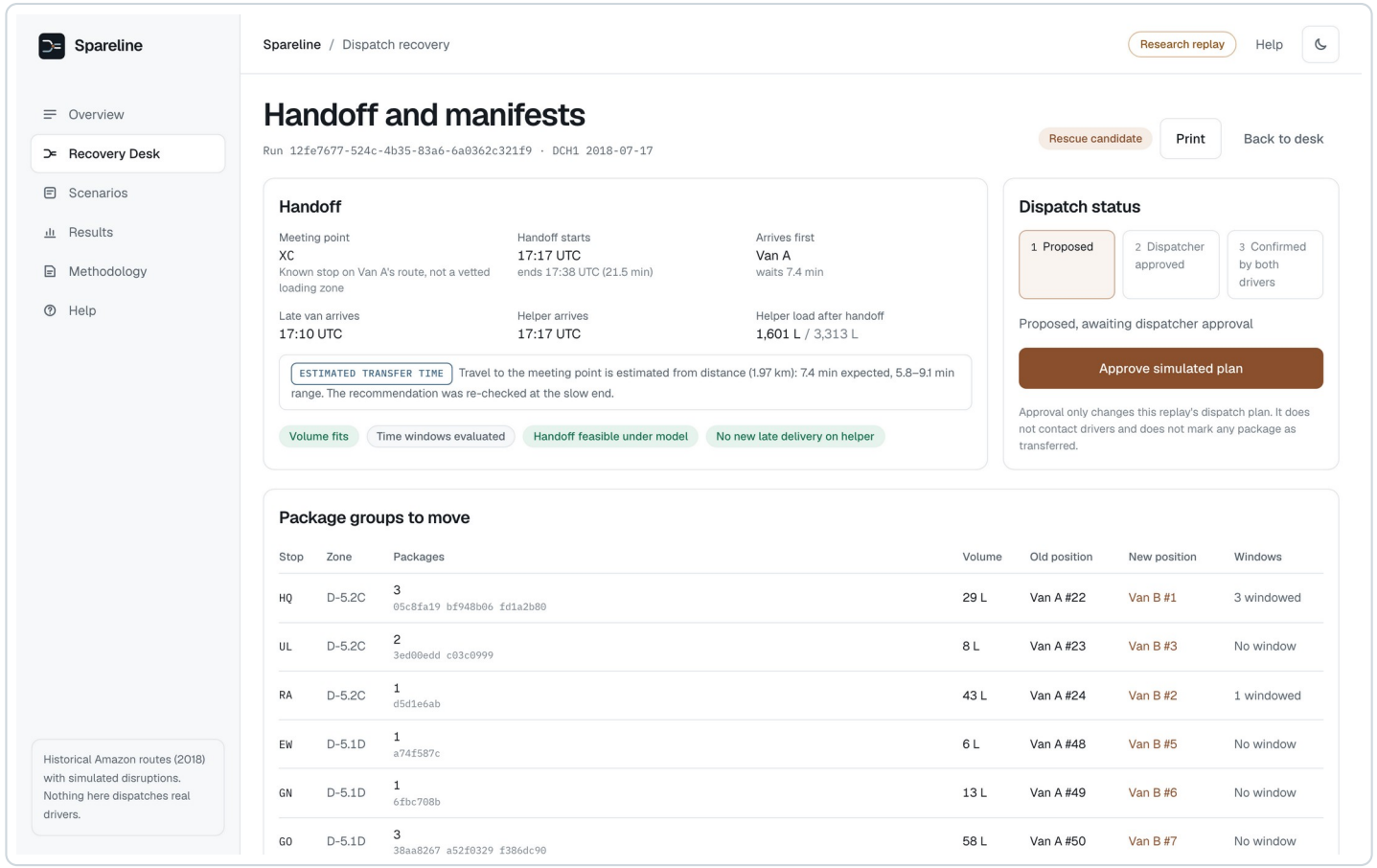
Doing nothing, nearest van and Spareline, with run ID and data source.

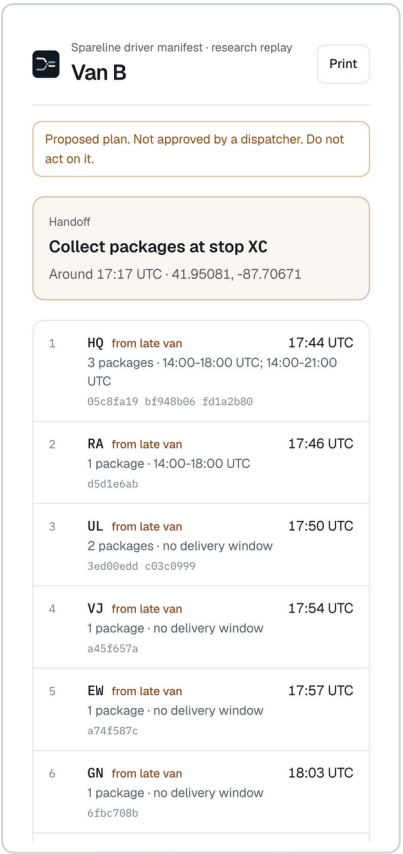
03 Labelled provenance

Historical, reconstructed, simulated, estimated and model results are tagged.

Inside one recommended rescue.

Scenario A in production: what the dispatcher approves and what the helper driver receives.





Meet at stop XC

Van A arrives first and waits; the handoff takes 21.5 min for 66 parcels.

30 stops move

Missed packages: 30 → 2. The helper gains no missed package.

Two-step sign-off

Dispatcher approves, then both drivers confirm. Only then is anything marked transferred.

Real historical routes. Explicitly simulated disruptions.

The source data is not a recording of vehicle failures, live GPS or actual rescue operations.

397

historical routes used, in 160 groups

54,405 stop records · 94,229 packages

Amazon / MIT Last Mile Routing Research Challenge, evaluation split (9,184 routes published in total)

2018 · RESEARCH DATA

Observed in the dataset

Stops, packages, time windows, capacity, recorded sequence and within-route travel times. Only 7.1% of packages carry a window; service time repeats per package, so a stop takes the max.

Constructed for evaluation

Same-depot, same-day groups of 2–4 routes; van progress reconstructed from each route's own matrix; between-route travel estimated (held-out MAE 60 s, q10–q90 coverage 81.1%).

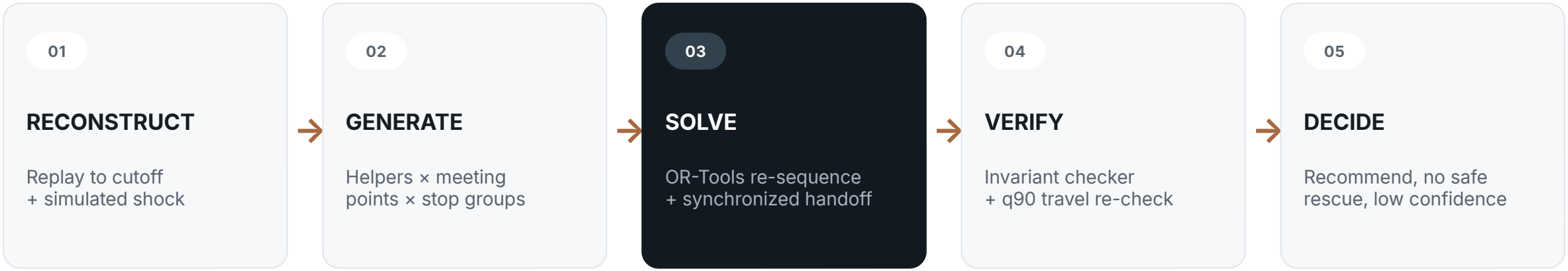
Simulated on purpose

Delays, area slowdowns and van failures, labelled wherever shown. 60-min shift limit and handoff time are declared assumptions.

Source: registry.opendata.aws/amazon-last-mile-challenges/ (evaluation split, downloaded 2026-11-10) • CC BY-NC 4.0; research use, not unrestricted commercial reuse.

A feasible plan is better than a confident guess.

A real optimization engine determines what can move, when vehicles can meet and what happens afterwards.



Optimization core

Both vans must reach the same stop location; the handoff starts when the later one arrives and lasts 5 min + 15 s per parcel. OR-Tools re-sequences the helper with soft window and shift penalties while a rank dimension keeps its own stops in observed order. An independent checker verifies conservation, capacity, synchronized arrival and forward time, then every cross-route leg is re-timed at its q90 estimate.

Objective: fewest missed packages on all vans → no new helper misses → fewest late minutes → least added driving and waiting.

Built, deployed and tested end to end.

What runs behind the demo, and how we know it works.

Browser	Next.js app on Vercel, deployed with the CLI. Map, decision, manifests, results.
API	FastAPI on our VPS in its own Docker project. Solver runs in worker processes; repeat requests reuse the same run.
Solver	Python + Google OR-Tools. Independent invariant checker on every plan.
Data	Postgres with versioned migrations: routes, stops, packages, travel matrices, runs, approvals, evaluation rows.

34 backend tests pass: data parsing, time zones, solver invariants, baselines, API flow, bad input

24 browser tests pass on the live site at 1440, 1024, 768 and 390 px

0 invariant violations in 456 evaluated plans

409 returned for a second approval, a stale revision or an export before approval

Approval state is kept per browser session, so judges never see each other's changes. Rate limits, input validation and CORS are on.

Same scenario. Three policies. Both vehicles scored.

The test is built to find where the rescue fails, not to produce a nice win rate.

01

Do nothing

Every van keeps its remaining observed order under the same disruption.

02

Nearest feasible helper

Closest van takes every at-risk stop at the late van's position by cheapest insertion; must fit capacity and shift.

03

Spareline

Optimize helper, meeting point, moved stops and order together; abstain when unsafe or uncertain.

- **456 predeclared scenarios, 114 held-out groups, 17 depots**
- **Measured: missed (late window or past shift), on-time windowed, helper harm**
- **4 shocks per group: 75 / 105 min delay, 1.6× slowdown, van failure; seed 2026**
- **Also: added drive and wait minutes, solver latency, abstentions, invariant errors**

Config fixed before the run (evaluations/config.json); 46 development groups used only for tuning and demo scenarios. Identical records, shocks, travel model and scoring for all policies; denominators and misses published.

Three scenarios. Three meaningful decisions.

Verified on the deployed application 2026-11-10. Each card is a live solver run; IDs shown for reproduction.

A

A rescue can help

DCH1, 2018-07-17 · 2 vans · donor 90 min behind (simulated)

scenario 8185981a · run 12fe7677

Recommended. Missed packages: 30 doing nothing, 19 nearest van, 2 Spareline. 30 stops move; helper gains 0 misses.

B

A rescue can hurt

DLA3, 2018-09-13 · 2 vans · donor 90 min behind (simulated)

scenario d711b3d2 · run 6c66370c

No safe rescue. Every transfer that helps the late van makes the helper late (HELPER_WOULD_BE_LATE).

C

No safe rescue exists

DSE4, 2018-07-05 · 2 vans · donor 90 min behind (simulated)

scenario 17d3d1ff · run 12abbae1

Low confidence. A transfer helps on mean travel but not with cross-route legs at q90; nothing is recommended.

Demo scenarios come from development groups and are not part of the aggregate results. Plans are advisory until a dispatcher approves and both drivers confirm.

What happened, including the misses.

Replay estimates under simulated disruptions · 456 scenarios on 114 held-out groups · eval_20261110T185425Z_21dfd2fe.

MEASURE	DO NOTHING	NEAREST VAN	SPARELINE	DENOMINATOR
Missed packages, all vans	24,474	24,889	**23,593**	195,279 remaining
Same, cross-route legs at q90		25,145	**23,713**	same cohort
On-time windowed packages	12,165	12,090	**12,200**	13,894 windowed
Helper packages newly missed (q90)		14 (129)	**0 (0)**	donor + helper scored
Better / tied / worse vs doing nothing			**74 / 382 / 0**	456 scenarios
Better / tied / worse vs nearest van			**102 / 350 / 4**	456 scenarios

881 missed packages avoided vs doing nothing; mean 1.932/scenario, 95% CI 1.474–2.434

74 rescues recommended; 280 of 354 at-risk scenarios abstained (76 low confidence)

3.65 s p95 solver time (mean 1.87 s); 0 invariant violations, 0 errors

Misses: the nearest van beat Spareline in 4 scenarios; in 2 it stayed ahead at q90, where the strict helper rule or calibrated-range rule withheld a transfer. Most misses come from the 60-min shift-limit assumption. Not field performance.

Built for a narrow, practical dispatch decision.

A research prototype can be useful before it is proven ready for live fleet operations.

Today: controlled research

Deployed replay of historical routes with clearly marked artificial disruptions, checkable outputs and published misses.

Next: operator pilot

With consent, compare recommendations against actual multi-vehicle telemetry and dispatcher outcomes.

Then: operational rollout

Add licensed fleet data, verified cross-route travel, human-confirmed transfers and monitored decision quality.

Known limits: obfuscated coordinates • no observed cross-route transfers • reconstructed, not GPS, progress • simulated shocks • assumed shift limit • no field validation • noncommercial licence

SPARELINE

Try the rescue decision.

The live demo runs the complete route-recovery loop: scenario, solver, decision, approval and manifests.

01 **Open Scenarios and run A, B or C**

02 **Change the disruption and recalculate**

03 **Approve, confirm both drivers, export**

Working demo URL

https://spareline-steel.vercel.app

Methodology: spareline-steel.vercel.app/app/methodology · Results: /app/results



Team: Timothy Popoola and Anu Kolapo

Research: Amazon/MIT Last Mile Routing Research Challenge (CC BY-NC 4.0), Merchán et al., Transportation Science 2022 • Google OR-Tools